

1. Administrative & Study Identification

Full Study Title: Study of Efficacy and Safety of Erenumab in Adult Chronic Migraine Patients **Official Title:** A 12-week Phase 3, Randomized, Double-blind, Placebo Controlled Study to Evaluate the Efficacy and Safety of Once Monthly Subcutaneous Erenumab 70 mg in Adult Chronic Migraine Patients **Short Title/Acronym:** DRAGON (CAMG334A2304) **Protocol Number:** CAMG334A2304 **NCT Number:** NCT03867201 **Posting ID:** 509831824415348270 **Sponsor Name:** Novartis **Company:** Novartis **Investigational Product (Drug Name):** Erenumab (AMG334) 70 mg **Phase of Development:** PHASE3 **Trial Status:** COMPLETED (Study has been completed) **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy and safety of once-monthly subcutaneous erenumab 70 mg compared with placebo for the prevention of chronic migraine. **Secondary Objectives:** To evaluate the achievement of $\geq 50\%$ reduction in monthly migraine days (MMD), change in monthly acute headache medication days, and change in migraine-related disability and productivity (mMIDAS scores).

2.2 Background & Rationale The purpose of this study was to evaluate the efficacy and safety of erenumab in patients with chronic migraine in the Asian population, a population not adequately represented in the global pivotal CM studies. Access to effective targeted preventive treatments like CGRP receptor antagonists remains a significant unmet medical need for patients experiencing debilitating chronic migraines in these regions.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** 1:1 (Erenumab 70 mg : Placebo) **Design Description:** This study used a single-cohort, 2-treatment arms, randomized, double-blind study design in adult subjects with chronic migraine.

3.2 Planned Enrollment & Duration Target N: 557 participants **Number of Sites:** 64 sites across 9 Asian countries/regions (Mainland China, India, Republic of Korea, Malaysia, Philippines, Singapore, Taiwan, Thailand, Vietnam) **Estimated Study Start:** August 2019 **Estimated Primary Completion:** August 2021

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Ages 18 to 65 years. 2. History of at least 5 attacks of migraine. 3. ≥ 15 headache days of which ≥ 8 headache days meet criteria as migraine days during the baseline period. 4. $\geq 80\%$ diary compliance during the baseline period.

4.2 Exclusion Criteria 1. Older than 50 years of age at migraine onset. 2. History of cluster or hemiplegic headache. 3. Evidence of seizure or major psychiatric disorder. 4. Cardiac or active hepatic disease. 5. Pregnant or nursing.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Erenumab 70 mg once monthly. **Route of Administration:** Subcutaneous injection. **Duration of Treatment:** 12-week double-blind treatment phase, followed by an open-label treatment period to last until end of PTA determined by the product launch in the country or the country's decision not to launch.

5.2 Concomitant & Prohibited Therapy Permitted: Regular acute migraine medications as approved by the trial doctor. **Prohibited:** Other monoclonal antibodies targeting the CGRP pathway or unapproved prophylactic migraine treatments.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -42 to -28	Informed Consent, Medical History, Eligibility Assessment (2 weeks)
Baseline	Day -28 to 0	4-week Baseline Diary, Randomization, First Dose
Interim	Week 4 & 8	Safety Assessment, Efficacy/Diary Review
End of Study (DB)	Week 12	Primary Endpoint Assessment, Transition to Open-Label
Follow-up	Week 24	Safety follow-up 12 weeks after the last DB treatment for subjects who discontinue or do not enter open-label

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Change From Baseline in Monthly Migraine Days During the Last 4 Weeks of the 12-week Treatment Period. A migraine day was defined as any calendar day in which the subject experienced a qualified migraine headache (onset, continuation, or recurrence). A qualified migraine headache was defined as a migraine with or without aura, lasting for ≥ 4 continuous hours, and meeting at least one of the following criteria: 1. ≥ 2 of the following pain features: Unilateral, Throbbing, Moderate to severe, Exacerbated with exercise/physical

activity. 2. %ge 1\$ of the following associated symptoms: Nausea and/or vomiting, Photophobia and phonophobia. **Measurement Method:** Electronic daily headache diary (patient-reported).

7.2 Secondary & Exploratory Endpoints - Change From Baseline in Migraine-related Disability and Productivity as Measured by the mMIDAS During the Last 4 Weeks of the 12-week Treatment Period. - Number of Participants With at Least 50% Reduction From Baseline in Monthly Migraine Days During the Last 4 Weeks of the 12-week Treatment Period. - Change From Baseline in Monthly Acute Headache Medication Days During the Last 4 Weeks of the 12-week Treatment Period.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Patient diary and HCP interviews assessed separately in the double-blind treatment period (DBTP) and the open-label treatment period (OLTP). **Serious Adverse Events (SAEs):** Standard SAE reporting within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Oversight by the sponsor's internal and external safety review boards.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) / Full Analysis Set. **Sample Size Justification:** Powered to detect a statistically significant difference in the primary endpoint of change from baseline in MMD between the erenumab 70 mg and placebo groups. **Handling of Missing Data:** Standard imputation methods for missing daily diary data.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring to assure %ge 80%\$ diary compliance and patient safety. **Audit Trail:** Data collection via standard eCRF systems with audit trails.

11. Study Identifiers & Governance

Primary ID: {{CAMG334A2304}} **Registry Number:** {{NCT03867201}}
Posting ID: {{509831824415348270}} **Sponsor/Lead Organization:** {{Novartis}} **Company:** {{Novartis}} **Study Title:** {{Study of Efficacy and Safety of Erenumab in Adult Chronic Migraine Patients}} **Official Regulatory Title:** {{A 12-week Phase 3, Randomized, Double-blind, Placebo Controlled Study to Evaluate

12. Clinical Framework (Protocol Specific)

Disease Area / Conditions: {{Migraine}} **Primary Condition:** {{Chronic Migraine}} **Preferred Name:** {{Migraine Disorders}} **Preferred UMLS Name:** {{Chronic Migraine}} **Semantic Type:** {{Disease or Syndrome}} **MeSH Code:** {{D008881}} **HPO Code:** {{HP: 0002076}} **SNOMED ID:** {{37796009}} **Drug Name / Drug Preferred Name:** {{erenumab}} **ATC Code:** {{N02CD01}} **Source Level:** {{5}} **Criteria Type:** {{PHASE_REQUIREMENT}} **Criteria Text:** {{Trial must be in PHASE3}} **Phase ID:** {{PHASE3}} **Status Name:** {{COMPLETED}} **Status Description:** {{Study has been completed}} **Diagnosis Threshold:** {{ \$\ge 15\$ headache days, \$\ge 8\$ migraine days/month}} **Medical Methodology:** {{CGRP receptor antagonism}} **Optimization Target:** {{Reduction in monthly migraine days (MMD)}}

13. Study Procedures & Methodology

Management Pathway: {{Standard clinical pathway for Migraine prevention}} **Education Protocol (HCP):** {{Diary training and administration protocol}} **Education Protocol (Patient):** {{Headache diary compliance training}} **Quality Audit Mechanism:** {{eDiary adherence tracking (\$\ge 80\%\$ required)}}

14. Observation & Follow-Up Schedule

Total Duration: {{12-week Double-blind + Open-label period}} **Onsite Visit Cadence:** {{Weeks 4, 8, 12, and Open Label visits}} **Remote Monitoring Frequency:** {{Continuous diary monitoring}} **Checklist Review Interval:** {{Every 4 weeks via diary}}

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[06-Apr-2026]				
Clinical Operations Lead	[Name]	_____	[06-Apr-2026]		Lead		
Statistician	[Name]	_____	[06-Apr-2026]				